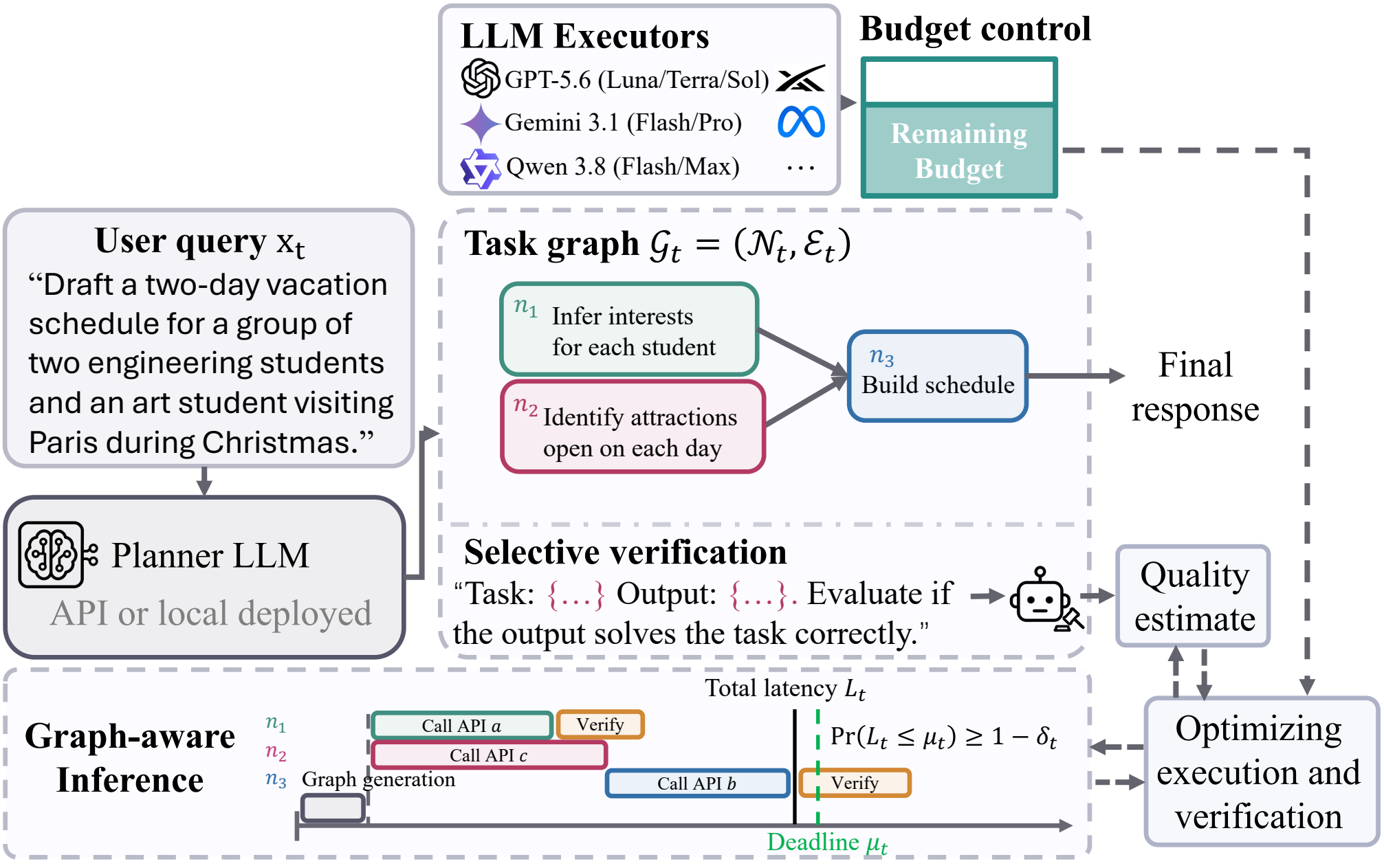

LLM Executors
GPT-5.6 (Luna/Terra/Sol)
Gemini 3.1 (Flash/Pro)
Qwen 3.8 (Flash/Max)
...
Budget control
Remaining Budget
User query x_t
"Draft a two-day vacation schedule for a group of two engineering students and an art student visiting Paris during Christmas."
Planner LLM
API or local deployed
Task graph $\mathcal{G}_t = (\mathcal{N}_t, \mathcal{E}_t)$
n_1 Infer interests for each student
n_2 Identify attractions open on each day
n_3 Build schedule
Final response
Selective verification
"Task: {...} Output: {...}. Evaluate if the output solves the task correctly."
Quality estimate
Graph-aware Inference
Total latency L_t
n_1
n_2
n_3
Call API a
Verify
Call API c
Graph generation
Call API b
Verify
$\Pr(L_t \leq \mu_t) \geq 1 - \delta_t$
Deadline μ_t
Optimizing execution and verification